An Image-based Method for the Morphological Analysis of Tendrils with 2D Piece-wise Clothoid Approximation Model

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Background

Tendrils are specialized, long, filiform organs in climbing plants (ex: *passionflower*), and sensitive to mechanical stimuli.
They can inspire and guide the design of slender robotic artefacts able to negotiate in complicated surrounding, for space application or environmental exploration.

Current challenge:

- Correctly describe and analyze the morphology of filiform structures
- Establish the relationship between morphological evolution and environmental triggering.

Approach and Main Contributions

- Image-based method with 2D piece-wise clothoid for curvature approximation, used to analyze the morphology of natural tendrils
- A sorting skeletonization algorithm able to handle abrupt changes in the direction of the extracted points
- Automatic method to identify the minimum number of 2D piece-wise clothoid spirals needed to represent a given tendril

Methodology

Step 1 — Image Pre-processing
Image resizing, distortion correction, binarization, Otsu's threshold segmentation and skeleton extraction

Step 2 — Weighted Sorting Skeletonization
Add tuning weight to adapt to the variation of curvature in generating direction

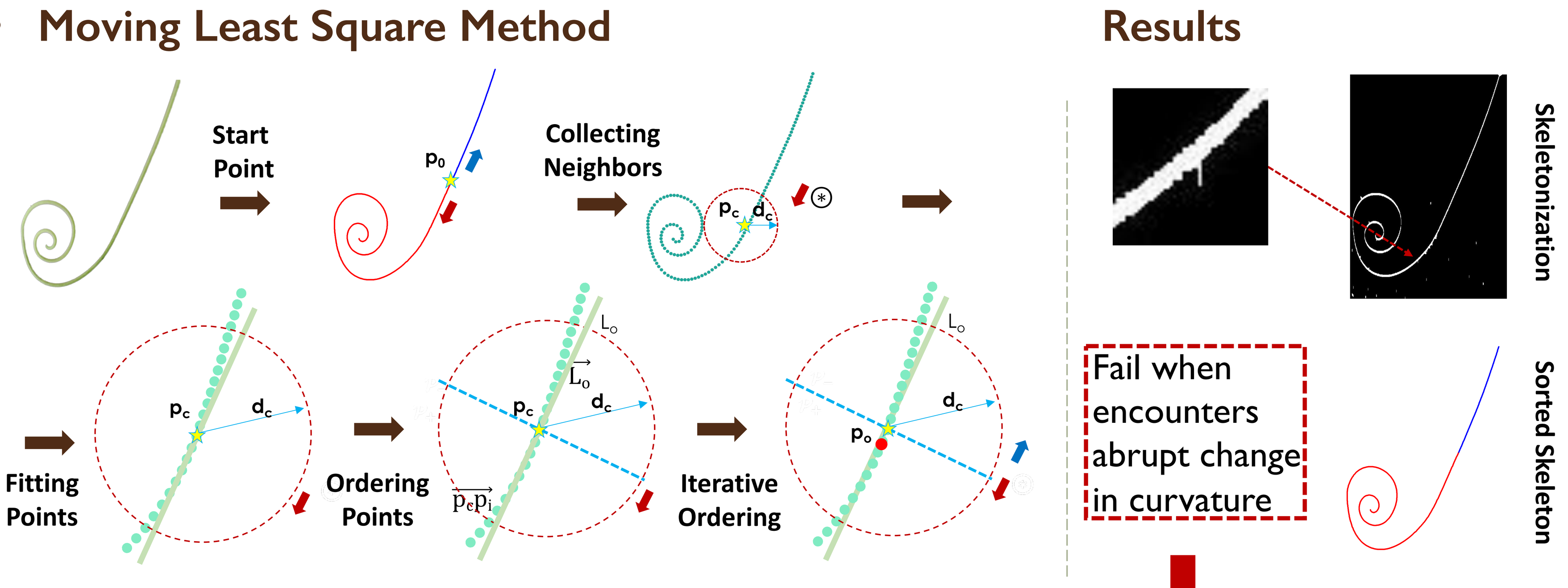
Step 3 — Discrete Curvature Estimation
Calculate circumcircle for each triplet at each current point
Choose a sliding window distance to assign the triplet (d = 50)

Step 4 — Automatic Selection of Segment Number
Find a minimal sequence of pieces in the curvature space (Dynamic Programming in a bottom-up fashion)

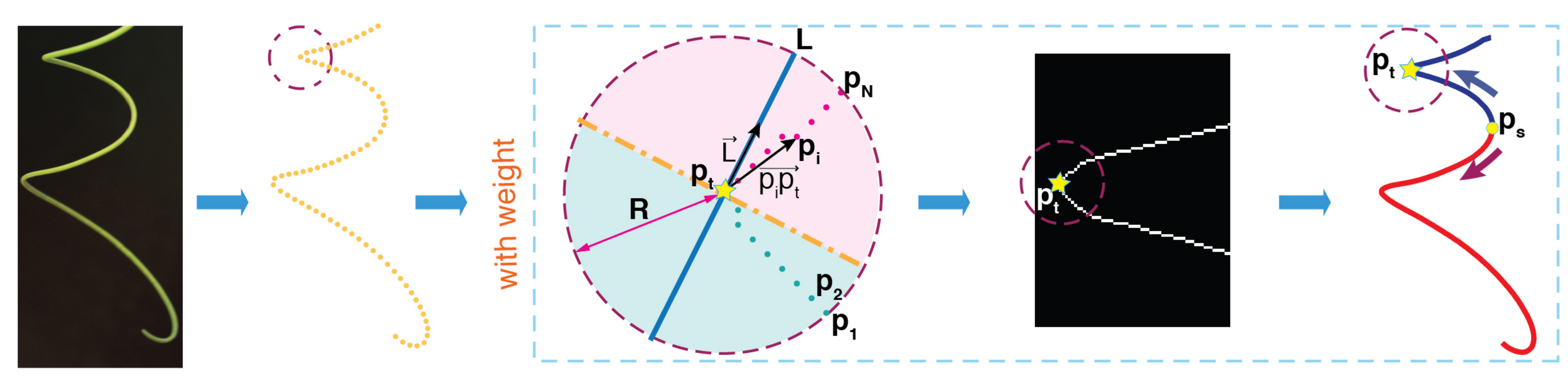
Step 5 — 2D Piece-wise Clothoid Representation
Parametrization of each clothoid piece by rational approximation of Fresnel integral
Transformation for curve alignment

Sorting Skeleton Points Algorithm – Step 2

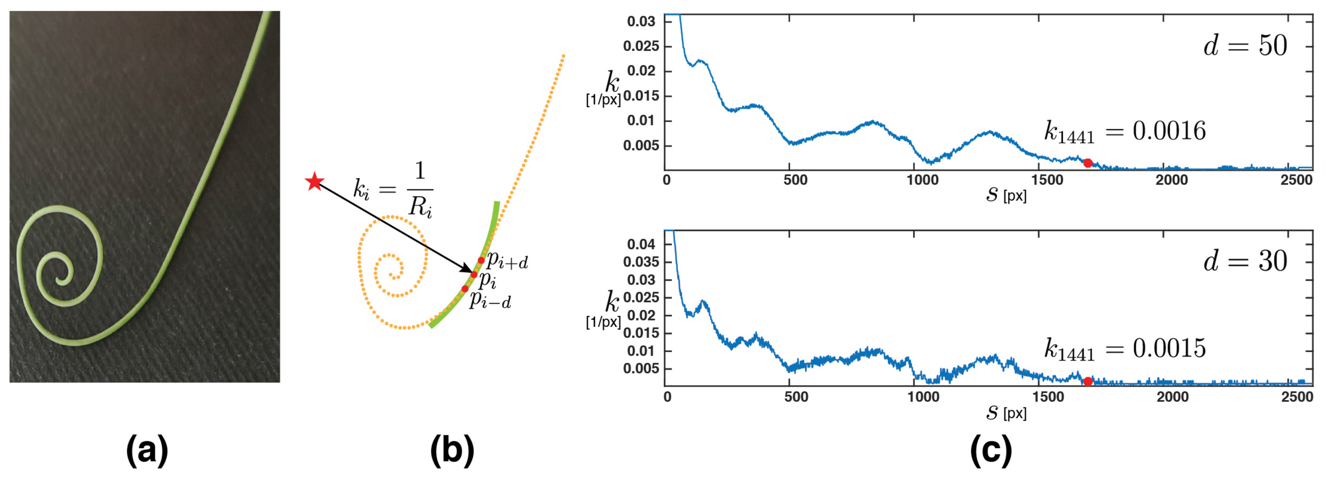
- Moving Least Square Method



- Weighted Moving Least Squares Method



Discrete Curvature Estimation - Step 3



Mapping from pixel-value space to arclength - curvature (s-k) space.

Sliding window distance affects the whole curvature smoothness.

Automatic Selection of Segment Number - Step 4

Dynamic Programming with an Up-sweep Approach

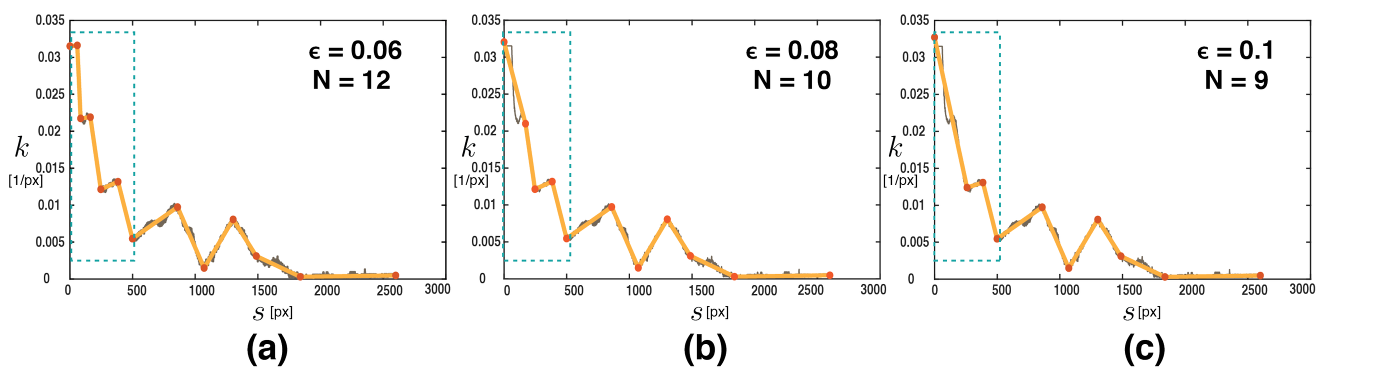
$$\mathcal{J}(p_i, p_j) = \min_{i<k<j} \{\mathcal{J}(p_i, p_k) + \mathcal{J}(p_k, p_j),\ \epsilon + \mathcal{E}_{fit}(p_i, p_j)\}$$

$$\mathcal{E}_{fit}(p_i, p_j) = \min_{a_{ij}, b_{ij}} \left(\sum_{t=i}^{j} |a_{ij} \cdot s_t + b_{ij} - \kappa_t|^2 \right)$$

$\mathcal{E}_{fit}$: the minimal sum of fitting errors caused by linear regression

a_{ij} and b_{ij} : slope and intercept of least-square fitting line

ϵ : penalty factor to the path treated as a single segment



2D Piece-wise Clothoid Representation - Step 5

2D Clothoid Spiral with Fresnel Integrals

$$\begin{pmatrix} x \\ y \end{pmatrix} = \pi B \begin{pmatrix} \mathcal{C}(l) \\ \mathcal{S}(l) \end{pmatrix}$$

$$\mathcal{C}(l) = \int_0^l \cos \frac{\pi}{2}\tau^2 \, d\tau \approx \frac{1}{2} - R(l) \sin \left(\frac{1}{2}\pi \left(A(l) - l^2 \right) \right)$$

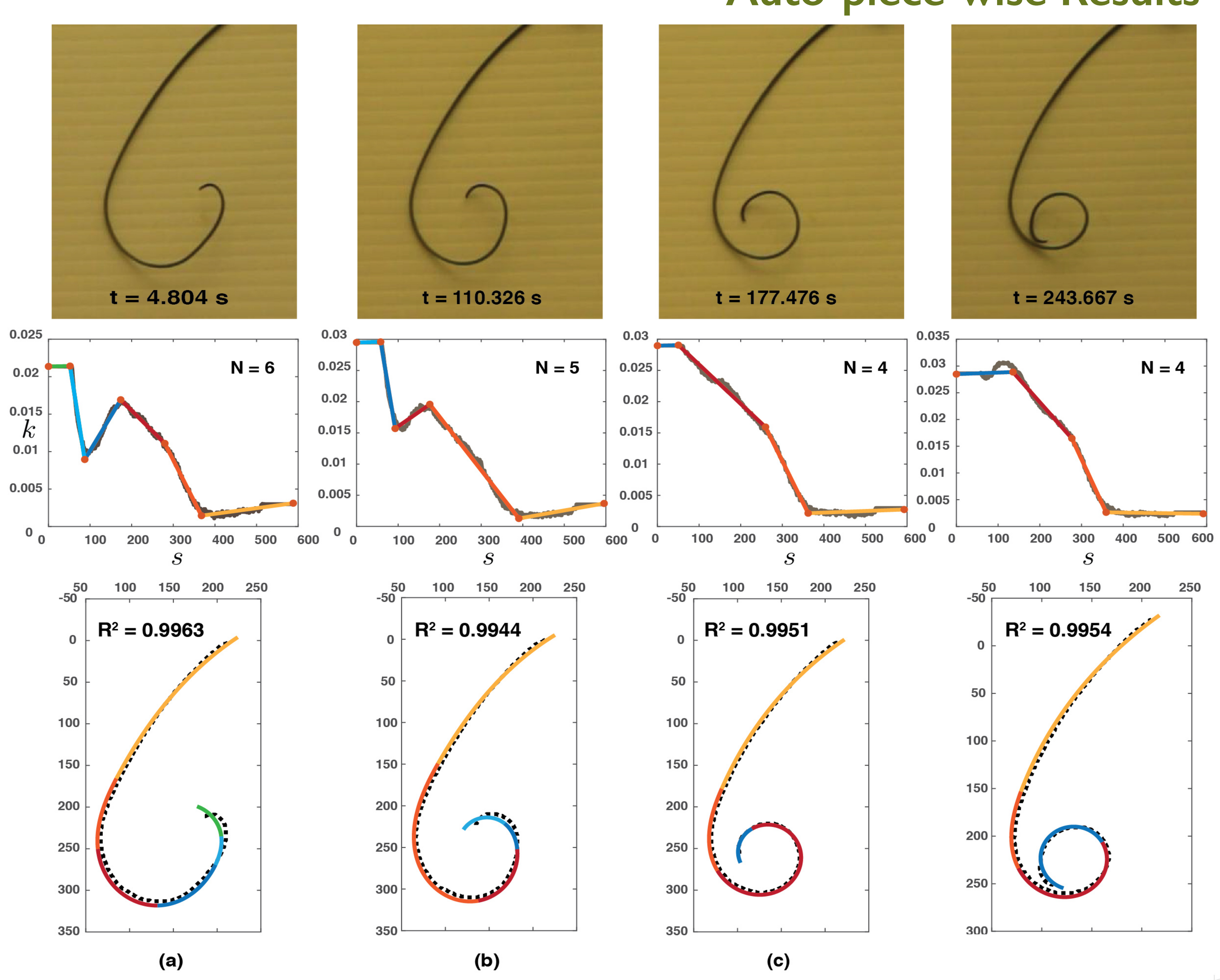
$$\mathcal{S}(l) = \int_0^l \sin \frac{\pi}{2}\tau^2 \, d\tau \approx \frac{1}{2} - R(l) \cos \left(\frac{1}{2}\pi \left(A(l) - l^2 \right) \right)$$

$$R(l) = \frac{0.506l + 1}{1.79l^2 + 2.054l + \sqrt{2}} \qquad A(l) = \frac{1}{0.803l^3 + 1.886l^2 + 2.524l2}$$

Rational Approximation with 1.7×10^{-3} error on R(l) and A(l)

Results

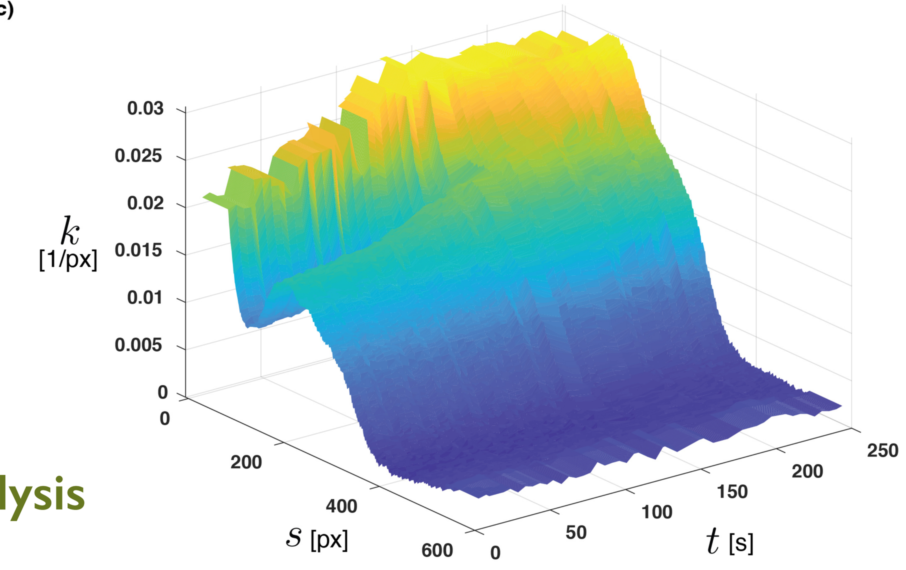
Auto-piece-wise Results



Supplementary video

We found that 4 to 6 segments were enough to represent curling shapes with high accuracy.

With this approach we can analyze the changes over time in the morphology of continuum adaptive structures.



Morphological analysis

Future Works

- Extend the model to 3D space to better analyze the shape and the behavior of tendrils
- Consider the strong self-occlusion of tendrils during their growing evolution in 3D space
- Analyze the morphological evolution of natural tendrils during support grasping
- Use the extracted feature to guide the design of tendril-like soft robots
- Investigating and developing control strategies for continuum growing robots

Contact

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Poster video

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